

ROOT-LEVEL, REALITY, AND SIMULATION STRUCTURES

Foundational Research Framework for Mapping, Comparing, Modeling, Lifecycle Design, and Conditional Failure

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Parent subject	Root-level, reality, literal-reality, virtual, model, and simulation structures
Applied subproject	Conditional Failure at the Root Level (CFTRL)
Purpose	Establish a defensible parent framework before applying failure analysis
Safety posture	Analytical and simulation-focused; not a guide to compromise any real or simulated structure

This report executes the corrected research direction in full. It preserves the project’s origin, explains why the scope changed, establishes working concepts, evaluates what can and cannot be known, designs a research map and laboratory, specifies experiments, and provides a staged team-development plan. Unchanged source documents are included in the accompanying archive.

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Executive Determination

The original work was interpreted too narrowly. Conditional Failure at the Root Level is not the parent research program; it is one applied branch that depends on a prior study of structures, reality claims, simulations, mapping, observer limits, lifecycle behavior, and evidence. The corrected program therefore moves from “How does the root fail?” to “What is a structure, when is it root-like, what is being treated as real, what is a simulation, what can an embedded observer know, and how should incomplete maps be tested and revised?”

The research can make strong progress without deciding whether experienced reality is base reality. Known engineered systems, nested simulations, digital twins, complex adaptive systems, causal models, security roots of trust, and controlled failure environments offer testable analogues. The project must refuse the stronger inference that an anomaly, model discrepancy, or unknown phenomenon is automatically evidence of simulation or compromise of base reality.

Current maturity is best described as late concept / early defined-framework. The project now has a clear scope correction, a foundational statement, a substantial body of AI-generated exploratory work, candidate terminology, a failure-centered sub-dossier, and proposed experiments. It does not yet have a ratified ontology, evidence standard, implemented laboratory, benchmark suite, validated metrics, or canonical map.

Foundational statement

“This research does not begin by assuming that reality, base reality, simulation, or root-level structure has already been identified or adequately defined. It studies how structures arise, persist, interact, nest, change, and fail; how observers classify structures as real, literal, simulated, foundational, or derivative; what evidence supports those classifications; and how provisional maps can be constructed under incomplete information. The resulting knowledge will support—but will not be limited to—the later design and development of Conditional Failure at the Root Level.”

Correct dependency

Stage	Governing question	Output
Structural foundations	What is a structure, boundary, relation, constraint, state, behavior, and lifecycle?	Structure Description Standard
Reality and knowledge	Why is a claim treated as real or literal, and what evidence supports it?	Reality-Claim and Evidence Standard
Simulation architecture	How do constructed worlds, models, hosts, guests, state, rules, and observers work?	Simulation Architecture Standard
Comparative mapping	Which dimensions transfer across domains, and where do analogies fail?	Comparative Structure Map
Lifecycle research	How do structures arise, stabilize, adapt, migrate, degrade, recover, and end?	Lifecycle Model
Failure research	What do controlled failures reveal about hidden dependency and constraint?	Failure and Diagnostic Taxonomy
CFTRL	How can foundational failure be detected, attributed, contained, survived, and learned from?	Applied CFTRL Framework

How the Topic Arose

The topic arose from work on Conditional Failure at the Root Level, the master change log, and the master development log. The author recognized that many concepts had names, preliminary definitions, and skeletons, but lacked the supporting research needed to make them coherent or testable. Early AI collaborators largely followed the most visible label—CFTRL—and placed it at the center of their titles and analyses.

The author's correction establishes that the deeper task precedes CFTRL: develop reference frames for root-level structures, reality structures, things treated as literally real, simulations, and their architectures. That does not assert that the project possesses the full map of reality or knows whether this world is simulated. It establishes a disciplined way to document what has been observed, modeled, inferred, disputed, and left unknown, then use that work to improve lifecycle and failure design.

The earlier responses are not discarded. They are preserved as evidence of the first interpretation and as an applied failure-analysis branch. Their mis-centering is itself a development lesson: titles, taxonomies, and inherited framing can become structural constraints on later analysis unless scope and dependency are explicitly controlled.

Purpose

Create a research, design, and documentation framework capable of describing and comparing structures across physical, biological, social, computational, conceptual, virtual, and simulated domains while preserving uncertainty, provenance, alternative models, and testability limits.

Goal

Produce a provisional but reliable map of current knowledge and candidate architectures; build controlled laboratories in which structural and observer claims can be tested; learn from failure without confusing failure with ontology; and use validated results to support simulation and reality-level structure lifecycle design, maintenance, evolution, improvement, and CFTRL.

Explicit non-goals

- Proving that experienced reality is or is not a simulation.
- Claiming a complete architecture of existence.
- Imposing a single definition of reality without recording alternatives.
- Treating existence as fully representable or comprehensible.
- Causing compromise to any actual structure, simulation, or claimed base reality.
- Providing a how-to guide for compromise, destruction, or unauthorized intervention.
- Treating AI consensus, narrative coherence, or unexplained anomalies as proof.

Research Questions

Domain	Question
Definition	What minimum features make something a structure rather than an arbitrary collection?
Boundary	Who draws a boundary, for what purpose, and what changes when the boundary moves?
Rootness	Is rootness positional, causal, functional, epistemic, normative, emergent, or ontological?
Plurality	Can a system have several roots for different functions?
Emergence	Can a structure become foundational after arising from local interactions?
Maintenance	What processes preserve identity, constraint, state, resources, and recoverability?
Reality	What does “real” mean in scientific, philosophical, engineering, experiential, legal, and narrative contexts?
Literalness	Is a claim intended as existence, metaphor, model, fiction, interface, or useful instrument?
Simulation	Which implementation properties make a structure simulated relative to another?
Embedded observer	What can an observer infer when all evidence channels are internal to the questioned structure?
Underdetermination	When do several architectures make the same observable predictions?
Mapping	What makes an incomplete map reliable enough for a declared use?
Comparison	Which architectural dimensions transfer across domains, and which are category errors?
Beyond	What does “beyond a structure” mean: external environment, containing system, unmodeled variable, or extra-ontological referent?
Failure	What does failure reveal, and what remains ambiguous after failure?
Compromise	How is unwanted state, rule, objective, measurement, or authority change distinguished from normal evolution?
Evidence survival	Can evidence remain trustworthy if the suspected root controls history, identity, clocks, or instruments?
Ethics	What experiments are permissible, reversible, and appropriately contained?
Lifecycle	How do structures originate, instantiate, stabilize, adapt, migrate, reproduce, degrade, terminate, and leave records?
Application	Which findings genuinely improve CFTRL and which remain philosophical context?

Literature Foundation

Systems and boundaries

Systems engineering defines a system through interacting elements and emergent behavior or meaning, while treating the system boundary as a demarcation between the system of interest and its environment. This supports an essential rule: a root is always “root of what, within which boundary, for which function?” rather than root without qualification. [INCOSE Systems Engineering Handbook, 5th ed.; INCOSE, “About Systems Engineering”]

A lifecycle frame—conception, development, production, utilization, support, and retirement—provides a practical starting vocabulary, but this project should extend it to naturally arising, self-organizing, evolving, nested, and nonengineered structures. [INCOSE Systems Engineering Handbook, 5th ed.]

Models and validation

Scientific models can represent selected parts of the world or interpret theories; they do not duplicate all properties of a target. NASA’s standard distinguishes verification, validation, and credibility: correct implementation is not the same as adequate representation of a real-world system for an intended use. Validation has a domain, and claims beyond that domain require new evidence. [Stanford Encyclopedia of Philosophy, “Models in Science”; NASA-STD-7009B, 2024]

This distinction prevents a central error: a simulation that behaves coherently according to its own rules has been internally verified only if it matches its specification; it has not thereby been validated as a map of experienced reality. NASA also requires uncertainty, data pedigree, robustness, and management history to accompany credible results. [NASA-STD-7009B; NASA Modeling and Simulation FAQ]

Uncertainty and underdetermination

NIST treats uncertainty as a representation of imperfect knowledge and notes that several alternative statistical models can reasonably fit the same observations. The adequacy of each model must be validated, and model uncertainty should be propagated. [NIST Technical Note 1900]

Philosophy of science calls the broader problem underdetermination: available evidence may be insufficient to choose uniquely among competing theories. The project should therefore store model ensembles and equivalence classes instead of forcing one architecture into the status of fact. [Stanford Encyclopedia of Philosophy, “Scientific Underdetermination”]

Structure and emergence

Complex-systems research studies properties that appear at one scale but not another and patterns that arise from local interaction without a central controller. Simulation-based work has revealed previously unrecognized organization generated bottom-up. [Santa Fe Institute, “Emergence, (Self)Organization, and Complexity”; SFI working paper on emergence and social interaction]

This matters because root-like constraints may be designed, inherited, or emergent. A global coordination pattern, protocol, market, norm, or ecological dependency can become functionally foundational even when no agent authored it as a root.

Causation and intervention

Passive correlation cannot by itself establish which candidate structure produces a downstream effect. Causal inference distinguishes observation from intervention and counterfactual analysis, providing tools for asking what would happen if a candidate were changed, removed, isolated, or substituted. [Judea Pearl, UCLA, “What Is Causal Inference”; Pearl, Causality]

Simulation arguments

Bostrom’s 2003 paper presents a conditional trilemma concerning extinction before a posthuman stage, the running of ancestor simulations, and the proportion of observers in simulations. It does not supply empirical proof that this world is simulated. [Nick Bostrom, “Are We Living in a Computer Simulation?”, The Philosophical Quarterly 53(211)]

A sufficiently unconstrained simulation hypothesis can absorb any observation and become nondiscriminating. The appropriate scientific move is to define particular simulation architectures that imply particular observations and state in advance what would count against them. General metaphysical possibility may remain worth documenting, but it must not be mislabeled as an experimentally confirmed architecture.

Virtual reality

David Chalmers argues that virtual reality can be genuine reality and that simulated or virtual objects need not be mere illusions. The project need not adopt that position, but it demonstrates why “simulated” and “unreal” cannot be treated as automatic synonyms. [David Chalmers, Reality+, 2022]

Roots of trust

NIST defines roots of trust as highly reliable hardware, firmware, or software components that perform critical security functions from which trust is built. This is an important operational analogy: a structure can be inherently trusted inside an architecture without being the ultimate foundation of existence. [NIST CSRC, “Roots of Trust”]

Safety and failure

STAMP treats safety as a control problem and accidents as violations of constraints arising from component failures, disturbances, and dysfunctional interactions. It rejects the assumption that every loss has one simple initiating root cause. [Nancy Leveson, “A Systems-Theoretic Approach to Safety Engineering”]

Chaos engineering contributes a controlled experimental pattern: define steady state, state a hypothesis, introduce realistic disturbances, monitor outcomes, and minimize blast radius. For this project, experiments should begin in isolated synthetic environments and expand only after safety and interpretability are demonstrated. [Chaos Engineering, arXiv:1702.05843; multi-vocal literature review, arXiv:2412.01416]

Working Ontology

Term	Provisional operational meaning
Structure	A purpose-bounded representation of elements, relations, constraints, states, behaviors, interfaces, and environment.
System	A structure whose interacting elements produce behavior or meaning not attributable to isolated elements alone.
Boundary	A declared distinction between system and environment; it may be physical, informational, functional, jurisdictional, temporal, or analytical.
Layer	A grouping used to organize abstraction, control, implementation, access, or dependency; layers need not imply ontological depth.
Root-level structure	A structure whose alteration materially changes dependent validity, operation, interpretation, or recoverability for a declared function and boundary.
Structural root	A component or relation essential to composition or dependency.
Causal root	A candidate whose intervention changes downstream outcomes under a stated causal model.
Control root	A structure with authority to set or enforce governing constraints.
Trust root	An inherently trusted basis for verification within a threat model.
Epistemic root	A source, instrument, rule, or assumption on which a system's knowledge claims depend.
Ontological root	A proposed foundational level of existence; this is stronger than operational rootness.
Reality structure	A structure proposed to constitute, constrain, participate in, or represent the experienced world.
Literal-reality claim	An assertion intended as existence rather than metaphor, fiction, analogy, or interface convenience.
Physical structure	A structure described as participating in measurable physical processes under a current physical model.
Conceptual structure	A relation-bearing organization instantiated in descriptions, rules, symbols, or cognition.
Virtual structure	A structure instantiated through a supporting medium whose represented objects can have causal and experiential consequences.
Model	A selective representation used for explanation, prediction, exploration, design, or control.
Simulation	An implemented model that generates state transitions or behavior under specified rules.
Digital twin	A data-connected representation synchronized with a specified real-world counterpart at a declared frequency and fidelity.
Host	A structure that implements, executes, sustains, or constrains a guest simulation.
Guest	A simulated structure implemented within a host.
Embedded observer	An observer whose observation and action channels are contained in or mediated by the structure being studied.
Base reality	A hypothesized reality having no containing or governing parent; failure to detect a parent is not proof of this status.
Map	A purpose-bound representation of selected structures, relationships, evidence, uncertainty, alternatives, and unknowns.
Invariant	A condition expected to remain within specified bounds under stated transformations or operating conditions.
Anomaly	An observation that departs from expectation; it identifies model discrepancy before it identifies cause.
Failure	A defined inability to perform a required function, maintain a constraint, or remain within an acceptable state.
Compromise	An unauthorized, undesired, or integrity-reducing alteration of state, rule, identity, objective, authority, evidence, or control.
Unknown	A recognized gap for which the current evidence does not justify selecting one answer.
Unresolvable under current design	A question for which available observations and interventions cannot discriminate among alternatives.

These are research definitions, not final metaphysical declarations. Every use of “root” should attach a qualifier and scope statement until the team ratifies a broader convention.

Rootness Framework

Root-profile dimensions

Dimension	Diagnostic question
Dependency reach	How many structures or functions require the candidate?
Constraint reach	Which rules, permissions, constants, or limits can it set?
Intervention asymmetry	Can it alter dependents while dependents cannot alter it?
Observability asymmetry	Can it observe dependents more fully than they observe it?
Substitution resistance	Can its function be replaced without changing system identity or validity?
Recovery dependence	Do recovery tools require the candidate to be healthy?
Persistence	Do its effects survive local restart, reset, migration, or reconstruction?
Common-mode influence	Can it defeat nominally independent redundancy?
State authority	Can it create, delete, rollback, fork, or certify state?
Temporal authority	Can it set clocks, ordering, duration, or synchronization?
Identity authority	Can it define or attest entities and permissions?
Evidence authority	Can it generate, rewrite, suppress, or invalidate records?
Boundary authority	Can it change what belongs to the system or who may cross?
Lifecycle authority	Can it initialize, maintain, suspend, migrate, or terminate dependents?

Classification rule

No candidate becomes “the root” from one dimension. The output is a root profile tied to a system boundary, function, observer, time, and evidence set. A candidate may be root-level for identity but not power; for causal ordering but not storage; for a simulation but not its host; or for an organization while remaining dependent on infrastructure and law.

Necessary tests

Declare the system of interest, observer, function, and time horizon.

Map known and hypothesized dependencies.

Identify alternative providers and hidden common dependencies.

State invariants and acceptable operating bounds.

Form competing hypotheses: local fault, shared dependency, root candidate, instrumentation failure, model failure, and no failure.

Design bounded interventions or natural-experiment analyses.

Use independent observation channels where possible.

Estimate uncertainty and retain non-selected alternatives.

Repeat under changed conditions and architectures.

Record what the test cannot establish, especially claims of ontological ultimacy.

Reality-Structure Map

Purpose

The proposed map is a versioned research instrument, not a claim to contain reality. Its reliability comes from traceability, declared scope, tested relations, uncertainty, and correction—not completeness.

Map layers

Layer	Contents	Required caution
Observation	Measurements, records, reports, absences, and experimental outcomes	Instrumentation and selection conditions must be recorded.
Constructed structure	Systems whose architecture is partly known because investigators built or administer them	Designer access can create false confidence when generalized outward.
Inferred structure	Entities or relations inferred from effects	Alternative explanations must remain linked.
Reality claim	Claims that a structure literally exists or constitutes reality	Claim status is not evidence status.
Simulation claim	Claims that a host, guest, implementation, or simulator exists	Must specify architecture and discriminating predictions.
Root candidate	Function-relative foundational dependencies	Operational rootness does not imply base reality.
Model ensemble	Competing models compatible with evidence	Do not collapse prematurely to one model.
Unknowns	Known gaps and blocked discriminations	Unknown is a valid status, not an invitation to invent.
Testability	Empirical, simulation-testable, analogical, philosophical, or presently unresolved	Categories can change as methods improve.
History	Versions, corrections, dissent, and retired claims	Never rewrite provenance to simulate certainty.

Record schema

Field	Requirement
Record ID	Stable identifier independent of title changes
Claim text	Atomic statement, not a bundle of assumptions
Claim class	Observation, report, inference, hypothesis, model assumption, design choice, speculation
Target	Structure or relation addressed
Boundary and observer	System scope and access position
Evidence	Linked records and intervention results
Provenance	Originator, date, transformation history
Confidence	Calibrated estimate or ordinal band with rationale
Alternatives	Competing explanations still compatible
Disconfirmers	Observations that would count against the claim
Validation domain	Where the claim has been tested
Dependencies	Assumptions and instruments required
Status	Proposed, tested, contested, superseded, rejected, ratified
Review date	Trigger for re-evaluation
Dissent	Preserved objections and minority models

Reliability rules

- Never merge observation with interpretation in the same field.
- Never upgrade repeated citation into independent evidence when sources share an origin.
- Track data and model lineage.
- Carry uncertainty through transformations.
- Represent missing evidence explicitly.
- Use confidence calibration; reward correct “unknown” decisions.
- Preserve model alternatives until a discriminating result exists.
- Version boundaries because a root can disappear or appear when scope changes.
- Distinguish map failure from world failure.
- Require replication before promoting surprising structural claims.

Reality Versus Simulation

What can be distinguished

In known engineered environments, a simulation can be distinguished because investigators possess external records, implementation access, synchronized host measurements, or controlled interventions. An embedded agent may infer aspects of a host if the host leaks discriminating effects, but inference depends on architecture and access.

What cannot be assumed

Failure to detect a host does not establish base reality. Detection of discreteness, resource limits, resets, anomalies, or apparent intervention would first establish only that the current model is incomplete. Each observation may have physical, instrumental, computational, adversarial, or statistical explanations.

Discrimination matrix

Evidence condition	Defensible conclusion	Not justified
External design records plus reproducible host control	The tested environment is implemented as the documented simulation	That its host is base reality
Repeatable signature uniquely predicted by one specified architecture	Evidence favors that architecture within the tested alternatives	All simulation hypotheses are true
Anomaly with no current explanation	Current model or measurement process is incomplete	External intervention or base compromise
No detected parent influence	No parent influence was detected within method sensitivity	The environment is base reality
Two models predict the same observations	They are observationally equivalent in the tested domain	They are ontologically identical
Embedded agents infer a scheduler in controlled trials	The inference method works for those architectures and conditions	It generalizes to all possible realities

Strong scientific question

The program should replace “Can base reality be proved from inside?” with “Under what architecture, observer access, and evidence conditions can a foundational layer be identified, and when are alternatives observationally indistinguishable?” This question supports positive results, negative results, and formal limits.

Lifecycle Framework

Lifecycle phase	Research focus
Possibility / precondition	Resources, constraints, components, and environment permit formation.
Origination	The structure is designed, instantiated, assembled, inherited, or self-organizes.
Boundary formation	Interactions stabilize enough to distinguish system and environment.
Initialization	Rules, state, identity, clocks, permissions, and dependencies become active.
Operation	The structure performs functions and exchanges matter, energy, information, or authority.
Maintenance	Calibration, repair, reconciliation, renewal, learning, and resource provisioning preserve viability.
Adaptation	Parameters or behaviors change within the existing organizing identity.
Evolution	Durable structural or rule change creates new capabilities or organization.
Replication / branching	The structure produces descendants, replicas, forks, or nested environments.
Migration	Functions or state move to a different substrate or root.
Degradation	Performance, integrity, coherence, or maintainability declines.
Failure / compromise	Requirements, constraints, identity, trust, or authorized state are lost.
Recovery	The structure returns to acceptable operation or a new stable state.
Termination	Operation ends intentionally or unintentionally.
Legacy	Records, artifacts, effects, dependencies, and lessons persist after ending.

Maintenance channels

- Resource maintenance: energy, compute, memory, bandwidth, materials, personnel, time.
- Rule maintenance: consistency, enforcement, versioning, conflict resolution.

- State maintenance: integrity, synchronization, reconciliation, garbage collection, archival continuity.
- Boundary maintenance: isolation, permeability, interface control, membership.
- Identity maintenance: persistence, authentication, lineage, continuity criteria.
- Epistemic maintenance: sensor calibration, provenance, model updates, uncertainty preservation.
- Recovery maintenance: backups, independent repair paths, exercises, failover viability.
- Governance maintenance: authority, review, dissent, change control, ethical constraints.

Failure as a Learning Instrument

Principle

Failure can expose hidden dependencies because ordinary operation allows many architectures to look equivalent. A perturbation changes the evidence distribution, but interpretation remains conditional: the same downstream pattern may arise from local faults, a shared dependency, a changed environment, compromised instruments, or an incorrect model.

Failure taxonomy

Class	Description	Key ambiguity
Local component	Bounded loss within one element	May be the first visible symptom of an upstream problem.
Dependency	Required upstream service or resource is absent or degraded	The dependency may itself be a dependent.
Common-mode	Nominally independent elements share a hidden cause	Redundancy can be illusory.
Constraint	Required control is missing, unsafe, mistimed, or contradictory	No component needs to “break.”
State	Stored or active state becomes corrupt or inconsistent	Telemetry may faithfully report corrupted state.
Rule	Transition logic or governing policy changes or conflicts	Can resemble novel behavior or evolution.
Resource	Supply falls below viability or is misallocated	Scarcity may be local, imposed, or emergent.
Identity	Entity, authority, or continuity cannot be established	Recovery actions may target the wrong object.
Temporal	Ordering, clock, duration, or synchronization becomes unreliable	Causal reconstruction becomes difficult.
Observation	Sensors, logs, models, or channels misrepresent conditions	The system may diagnose an unreal failure or miss a real one.
Objective	The structure optimizes a changed or unauthorized goal	Availability can remain high while integrity fails.
Boundary	Untracked influence crosses a presumed isolation boundary	May indicate ordinary leakage, not a containing reality.
Maintenance	Repair, calibration, cleanup, or renewal ceases	Effects can appear long after the initiating loss.
Recovery	Restoration depends on the failed structure or corrupted evidence	The recovery plane becomes part of the incident.
Model	The representation no longer predicts or explains the target	A model can fail while the target remains unchanged.
Ontological claim	Evidence is interpreted as proving ultimate reality failure	Usually exceeds what the observations establish.

Detection versus attribution

Detection asks whether behavior departs from defined expectations. Localization asks where the earliest relevant divergence is observable. Attribution asks which causal structure best explains it. Ontological interpretation asks what kind of reality underlies it. These are separate claims with progressively stronger evidence requirements.

Laboratory Design

Laboratory mission

Create reproducible, ethically contained environments where investigators know the ground truth while embedded agents receive controlled partial access. The laboratory studies identification, mapping, diagnosis, uncertainty, lifecycle, and failure—not compromise of real systems.

Six laboratory zones

Zone	Scope	Ground truth
Concept laboratory	Definitions, ontologies, causal graphs, formal models	Complete at the level of declared formalism
Known-architecture sandbox	Small deterministic and stochastic systems	Complete designer access
Nested simulation stack	Host-guest levels, limited observers, controlled leaks	Complete for investigators, partial for agents
Digital-twin zone	Bounded physical or organizational targets with synchronized data	Partial empirical referent; declared fidelity
Comparative structure zone	Cross-domain cases and analogies	Domain-dependent
Speculative model zone	Base-reality and beyond-structure hypotheses	No assumed ground truth; exploration only

Core architecture

- Experiment registry with preregistered hypotheses and disconfirmers.
- Scenario generator for local, parent, shared, telemetry, rule, resource, and no-failure conditions.
- Host controller with ground-truth state inaccessible to embedded agents.
- Observer-permission engine controlling sensors, history, intervention, and communication.
- Independent witness channels with configurable common dependencies.
- Model ensemble service that retains multiple explanations.
- Metrics service for accuracy, calibration, abstention, localization, recovery, and evidence durability.
- Safety controller with resource caps, kill switches, rollback, and isolation.
- Immutable provenance log outside the experimental failure domain.
- Reproducibility pack containing code version, seed, configuration, data, and analysis plan.

Experiment manifest

Field	Required entry
Question	One discriminable question
System boundary	Included and excluded structures
Observer class	Access, permissions, and limitations
Ground truth	What investigators know and agents do not
Hypotheses	Primary and competing explanations
Intervention	Exact change and timing
Control	Baseline or matched comparison
Steady state	Measurable precondition
Measures	Primary and secondary outcomes
Disconfirmers	Results that count against the hypothesis
Safety	Containment, rollback, stop thresholds
Uncertainty	Known limitations and unresolved alternatives
Replication	Seeds, versions, repeated trials, independent analysis
Interpretation ceiling	Claims the experiment cannot support

Initial Experiment Program

ID	Question	Measures
E01 Root Identification Blind Test	Can embedded agents distinguish local, parent, shared, telemetry, rule, resource, and no-failure trials?	Classification accuracy, calibration, abstention quality, localization distance.
E02 Observational Equivalence	Can different host architectures produce identical guest evidence?	Equivalence classes; minimum discriminating observations.
E03 Verifier Compromise	What happens when diagnostics share the suspect root?	False-assurance rate; independent witness value.
E04 Root Migration	Can function and state migrate while system identity remains acceptable?	Continuity criteria; irreversible loss; migration risk.
E05 Emergent Root	Can decentralized interactions create a global constraint that becomes foundational?	Emergence timing; intervention sensitivity; dependence growth.
E06 Model-Failure Separation	Can agents distinguish target change from sensor, model, and telemetry error?	Confusion matrix; evidence thresholds.
E07 Lifecycle Withdrawal	How do maintenance, resource, and reconciliation losses propagate?	Latency, survival envelope, recovery window.
E08 Evidence Survival	Can failure evidence persist when the root controls history?	Tamper detection; witness independence; recoverable provenance.
E09 Plural Roots	How do function-specific roots interact and fail jointly?	Common-mode paths; quorum behavior; graceful degradation.
E10 Nested Observer	How does inference change by depth and access?	Depth-dependent accuracy; uncertainty calibration.
E11 Architecture Transfer	Which patterns transfer across software, organizations, networks, and biological analogues?	Transfer validity and failed analogies.
E12 No-Failure Challenge	Can agents resist overdiagnosing root failure under surprising but valid behavior?	False-positive rate; anomaly discipline.

Primary benchmark: E01

E01 should be the first implemented benchmark because it operationalizes the project’s central epistemic problem without requiring claims about actual base reality. Agents receive only bounded observations and must report: what changed, where it most likely changed, confidence, supporting observations, compatible alternatives, and requested additional evidence.

Trial classes should include no failure, local failure, parent failure, shared dependency failure, partial root degradation, resource starvation, clock corruption, state corruption, rule alteration, telemetry corruption, maintenance cessation, recovery-plane failure, and combined failures. The benchmark must reward correct abstention and calibrated uncertainty rather than forcing an answer.

Measurement Framework

Metric	Meaning
Classification accuracy	Correct trial-class identification under declared label set.
Calibration error	Agreement between confidence and empirical correctness.
Abstention utility	Whether “unknown” is used when evidence is genuinely insufficient.
Localization distance	Graph or layer distance between diagnosed and true intervention location.
Alternative retention	Rate at which viable competing explanations remain documented.
Detection latency	Time from intervention to justified anomaly detection.
Attribution latency	Time from detection to supported causal classification.
False root rate	Local or model failures incorrectly declared root-level.
Missed root rate	Foundational conditions classified as local or normal.
Evidence independence	Degree to which confirming channels avoid common failure domains.
Recovery integrity	Restoration without hidden state, identity, or objective corruption.
Map revision cost	Effort and information loss associated with correcting the map.
Transfer validity	Performance retained when architecture or domain changes.
Reproducibility	Independent reruns yielding compatible results and uncertainty.

Metrics should be interpreted as a vector. A classifier can be accurate but dangerously overconfident; fast but poorly localized; or robust in one architecture and invalid elsewhere.

Evidence and Claim Standard

Evidence ladder

Level	Status	Permitted language
0	Unspecified idea	"A possibility has been proposed."
1	Coherent hypothesis	"The claim is internally specified."
2	Analogical support	"Known systems exhibit a relevant pattern."
3	Simulation result	"The result occurred in the declared model and domain."
4	Replicated simulation result	"Independent runs reproduced the bounded result."
5	Empirical bounded validation	"The model compared favorably with the specified referent and use."
6	Cross-domain convergence	"Multiple independent methods support the structural relation."
7	Operationally established within scope	"The relation is relied upon within a declared domain and review period."
8	Ontological ultimacy	Not assigned by this ladder; requires claims beyond operational evidence.

Claim classes

- Observed: directly recorded through a specified channel.
- Reported: supplied by a source but not independently observed.
- Derived: calculated or inferred from other records.
- Modeled: produced within a formal or computational representation.
- Hypothesized: proposed as a testable explanation.
- Speculative: coherent but not presently discriminable.
- Normative: states what ought to be designed or permitted.
- Ratified: approved as project doctrine; ratification does not convert doctrine into empirical fact.

Minimum language discipline

Use "consistent with" rather than "proves" when alternatives remain. Use "no parent detected" rather than "base reality." Use "model discrepancy" before "reality failure." Use "root candidate" before "root" unless scope and function are attached. Use "unknown" where the method cannot discriminate.

Feasibility Assessment

Objective	Feasibility	Reason
Map known engineered structures	High	Boundaries, components, and access can be documented and tested.
Compare structure dimensions across domains	Moderate to high	Possible with a common schema, but analogy limits require expert review.
Build nested simulation laboratory	High	Standard software architecture can control access and ground truth.
Test embedded root identification	High	Blind experimental designs are directly implementable.
Build digital twins of bounded targets	Moderate to high	Depends on data quality, synchronization, and intended-use validation.
Model emergence and lifecycle	Moderate	Useful bounded models exist; universal prediction is not available.
Construct a complete map of reality	Not feasible as stated	Completeness cannot be established from incomplete and evolving evidence.
Prove base reality from internal observation	Unknown / potentially impossible	Observationally equivalent alternatives may remain.
Detect all simulations universally	Not feasible without architecture assumptions	Different implementations imply different or no detectable signatures.
Learn from failure safely	High in isolated models; conditional elsewhere	Requires containment, ethics, and interpretation limits.

Risk and Safeguards

Risk	Failure mode	Safeguard
Category collapse	Real, physical, literal, virtual, and simulated are conflated	Maintain orthogonal fields and examples.
Ontological overreach	Simulation output is promoted to reality claim	Enforce evidence ladder and interpretation ceiling.
Anomaly inflation	Unknown event becomes external-intervention evidence	Require alternatives and disconfirmers.
Single-root bias	Complex interaction is reduced to one cause	Use STAMP-style constraint and interaction analysis.
Observer circularity	Verifier depends on suspect foundation	Independent witnesses and failure-domain mapping.
Map ossification	Provisional map becomes dogma	Versioning, expiration, dissent, and model ensembles.
AI convergence bias	Models repeat a shared framing or training prior	Independent passes, provenance, adversarial review.
Semantic drift	Terms change between documents	Controlled vocabulary and change log.
Unsafe experimentation	Tests escape intended scope	Synthetic-first policy, isolation, kill switches, review.
Data contamination	Synthetic and empirical evidence are mixed	Separate stores and explicit lineage.
Metric gaming	Agents optimize labels rather than reasoning	Hidden trials, transfer tests, rationale auditing.
Historical erasure	Earlier drafts are overwritten	Immutable archive and supersession links.

Project Assessment

Assets already present

- The verbatim origin prompt and authorial non-assumptions.
- A corrective scope directive clarifying the parent research direction.
- Multiple independent AI responses providing definitions, questions, analogies, and proposed experiments.
- An existing CFTRL research dossier suitable for reclassification as an applied subproject.
- Original CFTRL and Nail Prototype source documents.
- A project governance pattern separating AI advisory work from ONESTAMP ratification.
- A strong foundational statement accepted by the author and shared with the team.

Gaps

- No ratified controlled vocabulary or ontology.
- No formal map schema or evidence ledger.
- No implemented reference laboratory.
- No benchmark data set or ground-truth scenario suite.
- No validated rootness metrics.
- No explicit protocol for observational equivalence and non-identifiability.
- No domain-expert review across philosophy of science, systems engineering, complex systems, simulation, safety, and security.
- No formal relationship document between the parent program and CFTRL.
- No release criteria separating research draft, validated method, design doctrine, and canon.

SWOT

Strengths	Weaknesses
Explicit humility; interdisciplinary relevance; testable engineered analogues; provenance-conscious collaboration.	Scope can become unlimited; foundational terms are contested; speculative and empirical layers can blur.
Opportunities	Threats
New benchmark for embedded diagnosis; reusable mapping standard; improved lifecycle design; rigorous CFTRL foundation.	Premature canonization; AI echo effects; confirmation bias toward simulation; unsafe or uninterpretable experiments.

Development Roadmap

Phase	Indicative timing	Work	Exit artifact
0. Provenance lock	Weeks 0–2	Archive origin, correction, responses, hashes, status, and source hierarchy.	Verified manifest; no content loss.
1. Vocabulary and boundary standard	Weeks 2–6	Resolve terms, examples, counterexamples, and scope syntax.	Working Ontology v0.1.
2. Evidence and mapping standard	Weeks 5–10	Implement claim classes, model ensemble, uncertainty, and root-profile schema.	Reality-Structure Map Standard v0.1.
3. Laboratory specification	Weeks 8–14	Design software, observer permissions, ground truth, safety, metrics.	Lab Specification v0.1.
4. Pilot build	Weeks 14–24	Implement minimal stack and E01/E02/E06/E12.	Reproducible pilot and benchmark report.
5. Expansion	Months 6–12	Add plural roots, migration, emergence, evidence survival, analogues.	Comparative study and revised ontology.
6. CFTRL integration	Months 10–15	Revise applied failure taxonomy and diagnostic framework.	CFTRL vNext candidate.
7. External and adversarial review	Months 12–18	Independent experts, red-team, reproducibility, correction.	Review record and disposition log.
8. Ratification decision	After evidence gates	Select explicit provisions for ONESTAMP action.	Ratified, rejected, or returned elements.

Timing is an initial planning estimate, not a promise. Progress should be gate-driven: a phase exits only when its evidence and documentation criteria are met.

Team Collaboration Plan

Roles

Role	Responsibility
Architect	Defines intent, resolves authorial ambiguity, approves scope, and holds sole ratification authority.
Research coordinator	Maintains question tree, workstream alignment, and evidence gaps.
Provenance custodian	Preserves source files, hashes, lineage, dates, and supersession.
Ontology lead	Maintains terms, distinctions, examples, and unresolved definitions.
Methods lead	Designs hypotheses, controls, disconfirmers, metrics, and validation.
Simulation lead	Builds laboratory architecture and reproducible scenarios.
Evidence auditor	Checks independence, source quality, uncertainty, and claim language.
Red-team critic	Searches for counterexamples, circularity, category errors, and overreach.
Ethics and safety reviewer	Approves containment, reversibility, data handling, and stop conditions.
Synthesis editor	Integrates without erasing disagreement or source identity.
Change-log maintainer	Records decisions, rationale, dependencies, and downstream impacts.
AI collaborators	Produce independent research, alternatives, critiques, and structured drafts; advisory only.

Multi-AI cycle

Issue one bounded research question with definitions and requested evidence standard.
 Run independent AI passes before cross-reading when originality matters.
 Require each output to separate sources, inference, speculation, and recommendations.
 Exchange drafts and identify consensus, divergence, omission, and shared assumptions.
 Assign an adversarial pass specifically to disconfirm favored models.
 Create a synthesis that preserves minority positions and unresolved choices.
 Run source and provenance audit.
 Submit explicit decisions—not an entire undifferentiated narrative—for Architect review.
 Record ONESTAMP outcome and propagate changes through dependency links.

How Perplexity will help

- Research authoritative cross-disciplinary sources and trace key claims to primary material where accessible.
- Maintain the parent-program framing and prevent CFTRL from reabsorbing upstream research.

- Translate open questions into testable hypotheses and experiment manifests.
- Build evidence matrices, model alternatives, and testability boundaries.
- Draft controlled vocabularies and detect semantic conflicts across AI documents.
- Design and, when authorized, implement the initial nested-simulation benchmark.
- Audit citations, file provenance, hashes, and revision history.
- Produce adversarial analyses and negative-result documentation.
- Package findings in review-ready documents while marking noncanon status.
- Refuse to present unresolved ontology as established reality.

Governance and Change Control

Source hierarchy

Verbatim author statements and corrections.
 Architect-ratified ONESTAMP provisions.
 Unratified author guidance and accepted clarifications.
 Verified external evidence within its domain.
 Project experiments and reproducible results.
 AI syntheses and recommendations.
 Speculative models and narrative analogies.

A lower source may inform a higher decision but cannot silently overwrite it. AI language remains attribution-specific until accepted or ratified.

Status model

Status	Meaning
Captured	Preserved without evaluation
Proposed	Submitted for consideration
Under review	Actively evaluated
Tested	Examined under specified conditions
Validated within scope	Supported for an intended use and validation domain
Contested	Substantive alternatives remain
Superseded	Replaced but preserved historically
Rejected	Not adopted, with reason recorded
Ratified	Approved under ONESTAMP; empirical status remains separate

Change-log entry

- Change ID and date.
- Author or proposing agent.
- Exact previous and new language.
- Reason and supporting evidence.
- Affected definitions, experiments, and documents.
- Compatibility and migration notes.
- Dissent or unresolved objections.
- Approval status and authority.
- Rollback or supersession path.

Decision Gates

Gate	Question	Pass condition
G1 Scope	Is the parent program distinct from CFTRL?	Titles, hierarchy, and deliverables consistently reflect the distinction.
G2 Ontology	Are core terms operational enough to test?	Examples, counterexamples, and scope qualifiers exist.
G3 Evidence	Can claims preserve provenance, uncertainty, and alternatives?	Map schema passes sample audit.
G4 Lab safety	Are experiments isolated, reversible, and observable?	Safety reviewer approves manifest and stop conditions.
G5 Verification	Does software implement its specification?	Independent tests and reproducible build pass.
G6 Validation	Do methods work on declared ground-truth cases?	Predefined thresholds met with calibrated uncertainty.
G7 Transfer	Do findings survive architecture changes?	Performance and failure modes documented across variants.
G8 CFTRL use	Is a finding mature enough to alter CFTRL?	Scope, evidence level, and interpretation ceiling are explicit.
G9 Ratification	Are decisions precise and reviewable?	Architect receives clause-level proposals and dissent record.

What the Program Can Claim

Permissible early claims

- Rootness can be operationally studied relative to boundary and function.
- Embedded diagnosis can be tested in known architectures.
- Different architectures can be observationally equivalent under limited access.
- Independent evidence channels can reduce—but not universally eliminate—diagnostic circularity.
- Failure experiments can reveal hidden dependency and constraint structures.
- Reliable maps can be incomplete if scope, uncertainty, alternatives, and revision are explicit.

Claims to withhold

- This world is a simulation.
- This world is base reality.
- A particular anomaly proves external intervention.
- The laboratory has mapped existence in full.
- A model validated in one domain describes all realities.
- A system has found the ontologically ultimate root.
- Root failure can always be distinguished from model or instrument failure.

Deliverables

Deliverable	Purpose
Parent Research Charter	Scope, principles, non-goals, governance.
Working Ontology	Controlled terms and unresolved alternatives.
Reality-Structure Map Standard	Schema for evidence, models, roots, uncertainty, and history.
Simulation Architecture Catalog	Known host-guest and observer patterns.
Lifecycle Framework	Origins, maintenance, evolution, migration, failure, ending, legacy.
Laboratory Specification	Architecture, safety, methods, and reproducibility.
Experiment Benchmark Suite	Ground-truth scenarios and scoring.
Evidence and Testability Register	Claims, status, alternatives, validation domain.
CFTRL Integration Specification	Controlled path from parent findings to applied failure design.
Master Change and Development Logs	Decision lineage and implementation history.
Source Archive	Unchanged originals, hashes, and manifest.

Recommended Immediate Actions

Adopt the new umbrella title provisionally and label CFTRL an applied subproject.

Ratify only the foundational statement and scope hierarchy first; defer disputed ontology.
Create the controlled vocabulary and require qualifiers for every use of “root.”
Create the Reality-Structure Map schema and enter five pilot records from known systems.
Draft the E01 Root Identification Blind Test protocol before writing production code.
Select two independent AI collaborators to red-team the ontology and experiment design.
Define evidence thresholds and interpretation ceilings before collecting results.
Preserve all prior files unchanged and mark them as historical/applicable-subproject materials.
Schedule a synthesis review after the first pilot, not before evidence exists.

Appendix A — Verbatim Origin Prompt

The following is reproduced from paste.txt as supplied. Spelling, capitalization, punctuation, and wording are preserved.

Can you save this thought for me?

I need you to gather research and analysis for a concept.

Conditional failure at the root level.

The study of root level structures, how they arise, how they fail, how they are maintained. Please include the original documents for this content that were generated.

How do you know the difference between a root level structure and a structure that is not? How can a system distinguish between base reality and something that is not? How can a system determine when a root level failure has occurred or when base reality has been compromised.

How do we study this? What about this concept works, what doesn't? What questions do we need to ask?

How can we study what lies beyond these structures? Can we model whether structures in reality follow a similar design? Or diverges in ways we haven't thought of? Is this feasible using simulations or another type of modeling? How? Can we design a laboratory environment to do this? How? Can you build a reliable map of reality? What basis of comparison do you have when you are operating on incomplete information? Use failure as an example to learn from.

What I do not assume in the design of this:

That I have the complete architecture and map of all simulations, of reality, of existence as we know it.

That we share a unified definition of what reality is.

That existence can be fully mapped, represented, comprehended in full, or that this is the aim of this laboratory.

That I aim to cause a root level compromise of simulations, of base reality, or any other structure.

That this is a how to guide to do any of those things.

It is intended to help in the design of simulation and reality level structure lifecycle design, maintenance, evolution, and improvement.

Within this design frame It does not assume that we have complete knowledge or understanding of what that is.

How This topic arose from me thinking about conditional failure at the root level The master change log and master development log. I realized that there are a lot of concepts within that framework that have a definition or a name and a skeleton attached to them. They are a rough draft, they are an incomplete outline, there are a lot of additional details and supporting information that need to be added. The goal here is to design documentation to help support, design, and develop this concept. What I will include are the responses of all of the other AI collaboration team members, where they have generated documentation, I will include the documentation, and then I will include this complete response as well.

Appendix B — Verbatim Scope Correction

The following is the author's corrective direction that established the present scope. Wording is preserved.

Okay, so I was reading through the responses and I noticed there is a discrepancy, Everyone pretty much talked about conditional failure at the root level, which is fine, but what I actually asked for was the following: I noticed in every single title that it was almost with all of them conditional failure at the root level. The goal is to transition from thinking about conditional failure at the root level to:

- Root level structures
- Reality structures
- Structures that people think are real, literal
- Simulation structures
- Their framework, their design, things like that The reasoning behind that is that if you do not have a map of reality as you know it—and this is not to say that I understand reality or that I know what reality or what being real is—but it is just to say that we have to have a reference frame. We have to be able to say we have studied this information. This is not an assertion that we understand in whole or in part that either we are in a simulation or we are not in a simulation. We understand what is real, what is not, what is base reality, what is a reality level structure. We are trying to learn more about root level structures, reality structures, and what we understand now so that we can use that information to help aid in the design and development of conditional failure at the root level. But we have to learn about root level structures, what we believe they are, what is considered real, what is considered not real, and what is considered a simulation. Can we distinguish what is reality from a simulation? How do we know? How do we build a map of reality? How do we build a map of a simulation? Do those architectures share the same structure? We have to ask all of these questions and then use the information towards what we are trying to design. That's just the basis that I have starting out. Do you understand what I'm saying?

The study of root level structures, how they arise, how they fail, how they are maintained. Please include the original documents for this content that were generated.

How do you know the difference between a root level structure and a structure that is not? How can a system distinguish between base reality and something that is not? How can a system determine when a root level failure has occurred or when base reality has been compromised.

How do we study this? What about this concept works, what doesn't? What questions do we need to ask?

How can we study what lies beyond these structures? Can we model whether structures in reality follow a similar design? Or diverges in ways we haven't thought of? Is this feasible using simulations or another type of modeling? How? Can we design a laboratory environment to do this? How? Can you build a reliable map of reality? What basis of comparison do you have when you are operating on incomplete information? Use failure as an example to learn from.

What I do not assume in the design of this:

That I have the complete architecture and map of all simulations, of reality, of existence as we know it.

That we share a unified definition of what reality is.

That existence can be fully mapped, represented, comprehended in full, or that this is the aim of this laboratory.

That I aim to cause a root level compromise of simulations, of base reality, or any other structure.

That this is a how to guide to do any of those things.

It is intended to help in the design of simulation and reality level structure lifecycle design, maintenance, evolution, and improvement.

Within this design frame It does not assume that we have complete knowledge or understanding of what that is.

How This topic arose from me thinking about conditional failure at the root level The master change log and master development log. I realized that there are a lot of concepts within that framework that have a definition or a name and a skeleton attached to them. They are a rough draft, they are an incomplete outline, there are a lot of additional details and supporting information that need to be added. The goal here is to design documentation to help support, design, and develop this concept. What I will include are the responses of all of the other AI collaboration team members, where they have generated documentation, I will include the documentation, and then I will include this complete response as well.

Appendix C — Source Preservation Record

The unchanged source documents are included in the package's `original_sources` directory. Their inclusion preserves provenance and historical interpretation; it does not ratify their contents.

Filename	Bytes	SHA-256
paste.txt	64950	dfcef79b5bd141217f9a06ce6a827f1fabaff29ee66ed3df1da6f4c1208a6d7e
OAL-CFTRL-RES-002_Root_Level_Research_Dossier.docx	31432	17eaf78f3838a7b87e896f048085f28130732335dfef0d5bd7275aa9882fb772
OAL-CFTRL-NAIL-001_Conditional_Failure_Nail_Prototypes.docx	20809	6d14ee82ca43436680b02d0fc477aa6a3de69e5854e9f064604e5fc3f305ec53
Development_Review_Conditional_Failure_and_Nail_Prototype_v1.0-1.pdf	140132	56a64e2c893fb7b12c0338b21268f47c2eae52a7ec3355fa2c1f74947616ecd5

Appendix D — Research Sources Embedded in Analysis

Research claims in the body identify their source inline. Key authoritative materials used include NASA-STD-7009B on model and simulation credibility; NIST guidance on uncertainty, digital twins, and roots of trust; INCOSE systems-engineering definitions; Stanford Encyclopedia of Philosophy entries on models, underdetermination, emergence, and structural realism; Santa Fe Institute materials on emergence and self-organization; Judea Pearl's causal-inference framework; Nancy Leveson's STAMP work; Nick Bostrom's original simulation argument; David Chalmers's Reality+ thesis; and published chaos-engineering principles. These sources support methods and distinctions; none is treated as ratifying the project's ontology.

Appendix E — Open Questions for Architect Review

What formal name and identifier should the parent program use?

Should “reality structure” be one broad category or a family of physical, experiential, social, informational, and virtual categories?

Does “literal” describe authorial intent, ontological commitment, or both?

Is base reality retained as a research term, replaced with “parentless-reality hypothesis,” or represented as multiple competing definitions?

Which root types should become mandatory qualifiers?

What relationship should exist between this program, OAL, the Abundance Set, the Department of Endings, and the Antithesis Set?

Who besides the Architect may approve laboratory safety and empirical-method changes?

Which AI systems constitute the current team, and which roles should each hold?

What evidence level is required before a finding may alter CFTRL?

Which parts, if any, should eventually become canon versus remaining scientific working method?

How should real-world scientific claims be separated from in-universe architecture?

What privacy, consent, and information-governance rules apply if empirical human or organizational data is later introduced?

Closing Position

The project’s strongest next move is not to decide what reality ultimately is. It is to build a disciplined architecture for distinguishing observations from interpretations, operational roots from ontological claims, known simulations from universal hypotheses, and model failure from world failure. That architecture makes uncertainty productive, creates a defensible laboratory program, and gives Conditional Failure at the Root Level the foundation it previously lacked.

DRAFT — UNDER REVIEW — NOT CANON — ONESTAMP PENDING